

a Causal observed history



Frozen DINOv2



Past $a_{-1:0}$

Past-only context

Prefix $a_{1:h}$

Prefix GRU

Spatial encoder

FiLM

Temporal predictor

$H_h: 16 \times 96$

Dashed box:
frozen

b Anchor, mix and bound

H_h

$$Q_h = W_Q H_h$$

E_0

pre-FiLM

$$K_0 = W_K E_0$$

$$T_h = \text{softmax}_{\text{row}}(Q_h K_0^\top / \sqrt{96} + 4I)$$



$\times$

H_h

Patch gate

$$g_h = \sigma(W_g H_h + b_g)$$

Fixed Z_0



$1 - g_h$

g_h

Σ

$+$

$\hat{Z}_h$

project

H_h

R

tanh bound

-1

Δ_h

+1